

Jida Wang

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Education

Nanyang Technological University, School of Electrical and Electronic Engineering **Singapore**
M.Sc in Power Engineering *Aug. 2025 – 2027*

- **CGPA(until now):** 4.00/5.00
- **Thesis:** Cyber-Resilient Learning-Based Forecasting of Renewable Energy Generation in Multi-Plant PV and Wind Farms

Beijing Institute of Technology, School of Automation **Beijing, China**
B.Sc in Electrical Engineering and Automation *Sept. 2021 – June. 2025*

- **Average Scores:** 82.89/100
- **Thesis:** Optimized Scheduling Strategy for Power Grids Considering Complementary Characteristics of Renewable Energy Forecast Errors

Publications

A Mamba-Based Denoising Autoencoder for Stealthy FDIA Detection in Power Systems(Accepted) *August. 2026*

Jida Wang, Ziming Yan

2026 IEEE 18th Asia-Pacific Power and Energy Engineering Conference(APPEEC).

Optimized Scheduling Strategy for Power Grids Considering Complementary Characteristics of Renewable Energy Forecast Errors *July. 2025*

Jida Wang, Wenshuang Liu, Jie Yang, Xi Zhang, Zi'an Li, Yingli Huo

2025 5th IEEE International Conference on Industrial Electronics for Sustainable Energy Systems(IESES).

Research Experience

Lab of Dr. Yan Xu, Nanyang Technological University, Singapore *Sep. 2025 - Now*
Research Assistant: Resilience Enhancement in Sustainable Power Grids via Mamba-Driven Anomaly Detection Advisor: Dr. Yan Xu

- Engineered a cyber-physical time-series simulation environment for the IEEE 39-bus system using Python and Pandapower, integrating realistic 500 MW wind and solar generation profiles.
- Formulated non-linear AC Optimal Power Flow (AC-OPF) utilising the IPOPT solver to establish physical operational baselines, incorporating custom structural constraints for inverter-based resources.
- Designed and simulated coordinated cyber-threat models, encompassing workload FDIA and targeted SCADA sensor spoofing, to systematically evaluate modern grid vulnerabilities.
- Developed an end-to-end Mamba-driven State-Space anomaly detection framework with linear-time sequence modelling capabilities to efficiently process high-frequency multisensor time series.
- Implemented a Denoising Autoencoder (DAE) Fast Online Calibration strategy with sliding windows to continuously learn the dynamic healthy grid manifold and prevent Identity Mapping Collapse.
- Achieved accurate spatio-temporal localisation of stealthy sensor attacks with zero temporal hysteresis and a 20x to 30x Signal-to-Noise Ratio (SNR) gain.
- Publishing a paper with the title of A Mamba-Based Denoising Autoencoder for Stealthy FDIA Detection in Power Systems on 2026 IEEE 18th Asia-Pacific Power and Energy Engineering Conference(PES APPEEC26).

Lab of Dr. Yan Xu, Nanyang Technological University, Singapore *Sep. 2025 - Now*
Research Assistant: Advanced Energy Management and Data Fusion for Smart Mi- Advisor: Dr. Yan Xu

crogrids and Smart Nanogrids (Joint R&D Project with LITE-ON)

- Applied YOLOv8 and OCR techniques to identify equipment and extract labels from SLDs, with adaptive handling of diverse symbol standards and drawing conventions. Helped to build a symbol-to-parameter mapping library and combined deep learning with a rule-based engine for recognition.
- Helped to establish bidirectional mapping between power flow models and SLD images, supporting both reverse engineering from images to models and generation of standardised SLDs from models, enabling integrated topology visualisation and dynamic power flow analysis.
- Integrated physical power flow computation results with SLD visual features via a multi-modal Transformer framework to enhance grid state awareness and forecasting capability. Designed the system to support rapid modelling for complex industrial distribution networks and renewable integration scenarios.
- Helped to develop an adaptive electrical single-line diagram (SLD) information extraction and multi-modal power flow modelling system, enabling automatic device and connection data extraction from SLDs to construct corresponding power flow analysis models.

Renewable Power Grid Analysis, Optimisation and AI Application Group, Beijing Institute of Technology

Nov. 2024 - June. 2025

Advisor: Dr. Xi Zhang

Research Assistant

- Proposed and designed a hot reserve constraint that explicitly incorporates the complementary characteristics of renewable energy output forecast errors using robust optimisation.
- Developed an optimised power grid scheduling model considering the complementary characteristics of renewable energy forecast errors.
- Designed a blended algorithm to solve the model.
- Published a paper with the title of Optimized Scheduling Strategy for Power Grids Considering Complementary Characteristics of Renewable Energy Forecast Errors on 2025 5th IEEE International Conference on Industrial Electronics for Sustainable Energy Systems(IESES25).

Electrical Engineering and Control Lab, Beijing Institute of Technology

Mar. 2024 - July. 2024

Advisor: Prof. Xiwei Peng

Research Assistant

- Completed the control of the lift following the implementation process of a real engineering project, with a focus on the lift industry.
- Designed a control algorithm using the Siemens S7-1200 series PLC as the control system, with the lift simulation model as the controlled object.
- Participated in the 18th Siemens Cup CIMC China Intelligent Manufacturing Challenge in the Intelligent Manufacturing Engineering Design and Application category, focusing on Discrete Industry Automation - Logic Algorithms and earned provincial second prize.

Cambridge AI Lab, remotely at Cambridge University

Jan. 2023 - March. 2023

Advisor: Prof. José

Miguel Hernández-Labado

Research Assistant

- Presented the YOLOv5 algorithm for object detection tasks in highway safety, achieving high accuracy through the use of over 20000 annotated samples for training, validation, and testing, whose algorithm consists of a feature extraction layer, feature integration layer, and detection layer, and is evaluated using precision-recall curves and confusion matrices.

Software Development

Software Systems

Sep. 2025 - Now

- Single-Line-Diagram-Driven Multimodal Distribution Network Digital Twin Automation Platform

Control Systems

Mar. 2024 - May. 2024

- Designed a two-story building lift system using a TL082 integrated operational amplifier and 74LS00, incorporating necessary RC circuits for generating square waves, narrow rectangular pulse signals with 5%-35% duty cycle, and sine and cosine signals, emphasizing state machine (FSM) and datapath design methods to improve system efficiency and stability.
- Designed and implemented a flywheel speed control system for spacecraft simulation using a DC motor and PWM, achieving zero steady-state error, less than 10% speed overshoot, and less than 5% current overshoot,

with verified dynamic performance and stability through simulation, while also addressing economic and environmental considerations.

Awards

Third Class Scholarship , Beijing Institute of Technology	<i>2022, 2023</i>
Second Class Scholarship , Beijing Institute of Technology	<i>2024</i>
Provincial Second Prize , 18th Siemens Cup CIMC China Intelligent Manufacturing Challenge in the Intelligent Manufacturing Engineering Design and Application category, focusing on Discrete Industry Automation - Logic Algorithms	<i>2024</i>

Technologies

Languages: Mandarin Chinese (Native Speaker), English (IELTS CD 8.0, R9.0 / L8.5 / S6.5 / W7.0), Spanish (Elementary), Japanese (Elementary)

Programming Languages: MATLAB, Python

Library/Framework/Tool/Software: Pytorch, Simulink, PLC, SolidWorks, Altium Designer, Julia